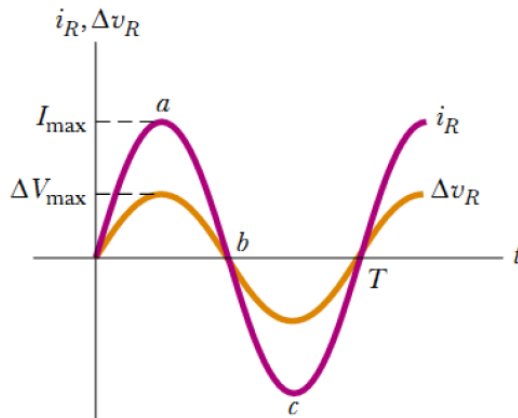


Alternating Current

Resistors in an AC Circuit



Plots of the instantaneous current i_R and instantaneous voltage Δv_R across a resistor as functions of time. The current is in phase with the voltage, which means that the current is zero when the voltage is zero, maximum when the voltage is maximum, and minimum when the voltage is minimum. At time $\tau=T$, one cycle of the time-varying voltage and current has been completed.

The average power delivered to a resistor that carries an alternating current is

$$\mathcal{P}_{av} = I_{rms}^2 R$$

The root-mean-square, rms current.

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 0.707 I_{max}$$

$$\Delta V_{rms} = \frac{\Delta V_{max}}{\sqrt{2}} = 0.707 \Delta V_{max}$$

- Example: What Is the rms Current?**

The voltage output of an AC source is given by the expression $\Delta v = (200 \text{ V}) \sin \omega t$. Find the rms current in the circuit when this source is connected to a $100\text{-}\Omega$ resistor.

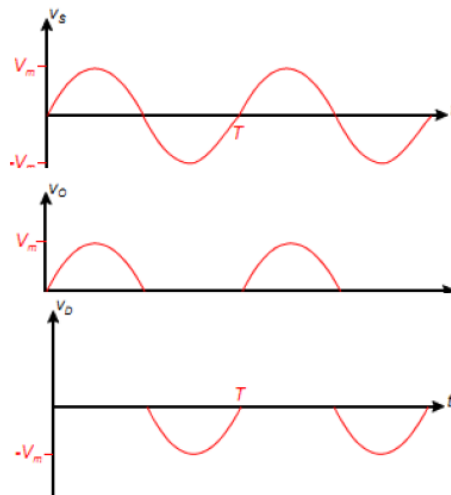
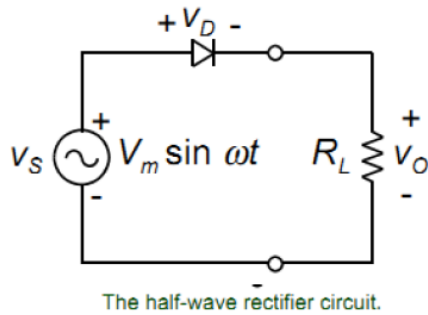
Solution Comparing this expression for voltage output with the general form $\Delta v = \Delta V_{max} \sin \omega t$, we see that $\Delta V_{max} = 200 \text{ V}$. Thus, the rms voltage is

$$\Delta V_{rms} = \frac{\Delta V_{max}}{\sqrt{2}} = \frac{200 \text{ V}}{\sqrt{2}} = 141 \text{ V}$$

Therefore,

$$I_{rms} = \frac{\Delta V_{rms}}{R} = \frac{141 \text{ V}}{100 \Omega} = 1.41 \text{ A}$$

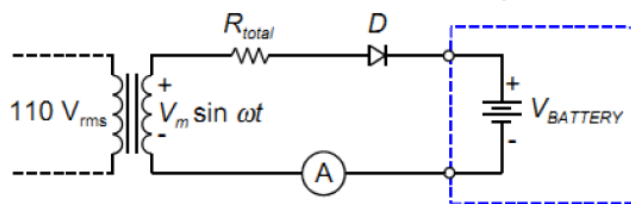
Diode Applications - The Half-Wave Rectifier



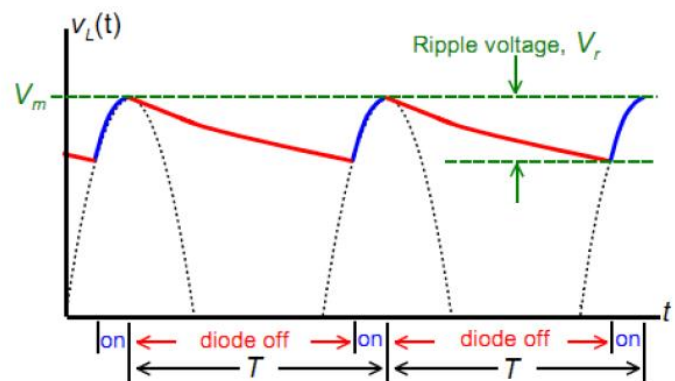
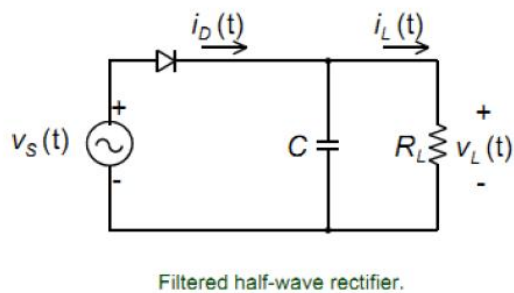
Waveform of voltage source.

Output voltage waveform.

Diode voltage waveform.

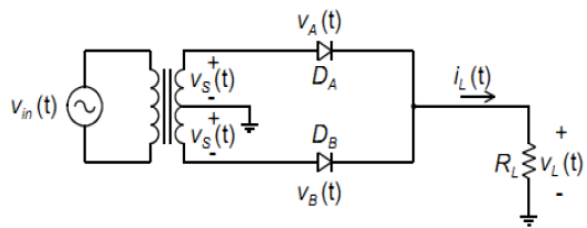


A Typical Battery Charging Circuit

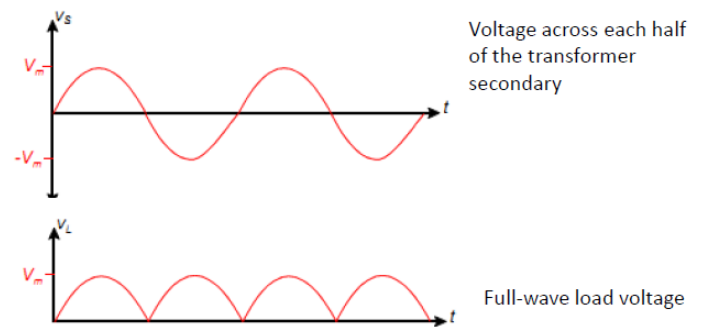


Load voltage waveform in the filtered half-wave rectifier.

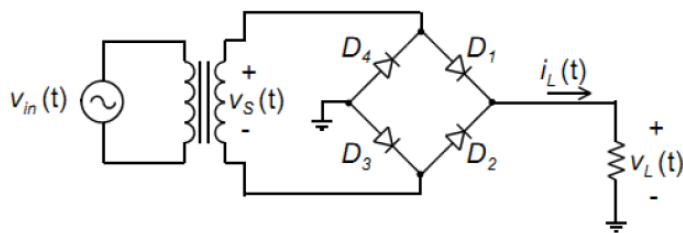
Diode Applications - The Full-Wave Rectifier



The full-wave rectifier

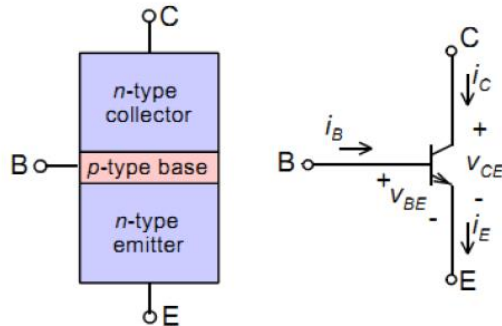


Diode Applications - The Bridge Rectifier



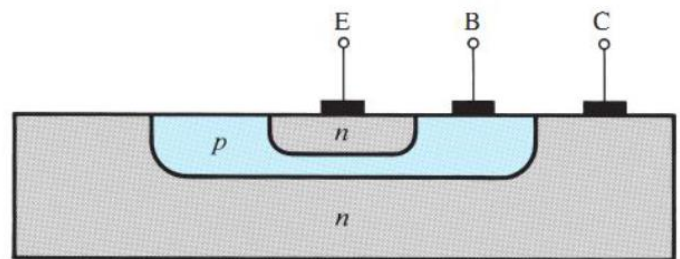
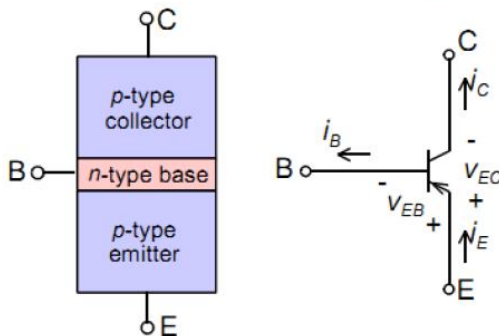
BJT Transistors

A npn BJT and its schematic symbol.



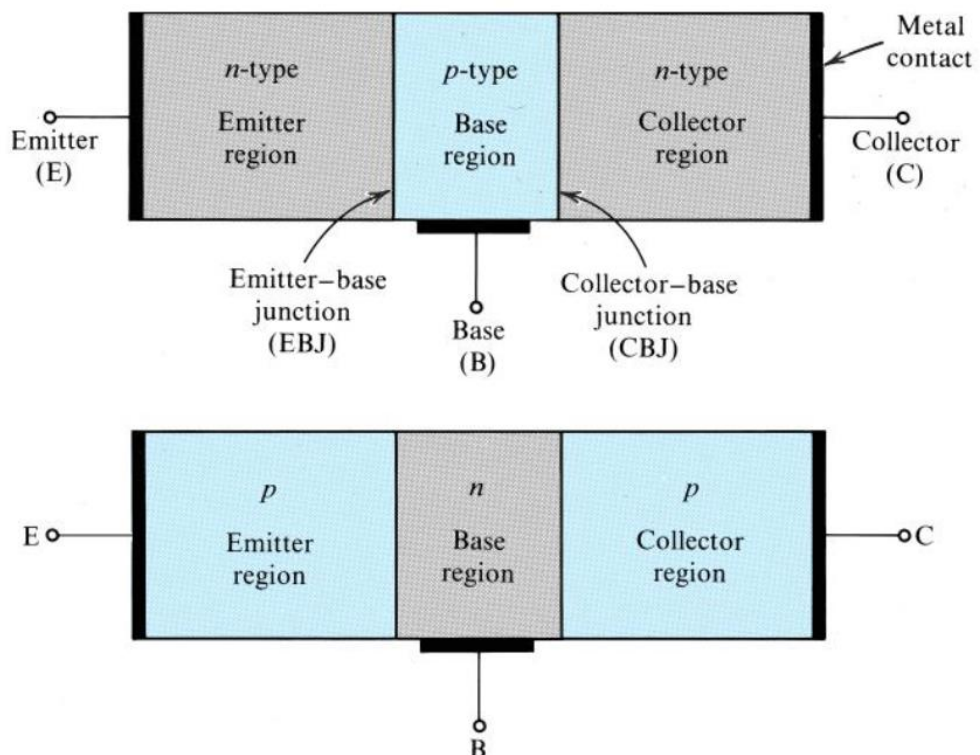
The BJT is a nonlinear, 3-Terminal device based on the junction diode. A representative structure sandwiches one semiconductor type between layers of the opposite type. We first examine the npn BJT

A pnp BJT and its schematic symbol.



Cross-section of an npn BJT.

The npn Transistor



A junction transistor consists of a semiconductor material in which a layer of n-type is sandwiched between two layers of the p-type semiconductor. Alternatively a transistor may consist of a layer of p-type between two layers of n-type materials. The semiconductor sandwich is extremely small. The three portions of a transistor are known as emitter, base and collector. The arrow on the emitter lead specifies the current flow when the emitter-base junction is biased in the forward direction. The symbols V_{EB} , V_{CB} and V_{CE} are the emitter-base, collector-base and collector-emitter voltages respectively.

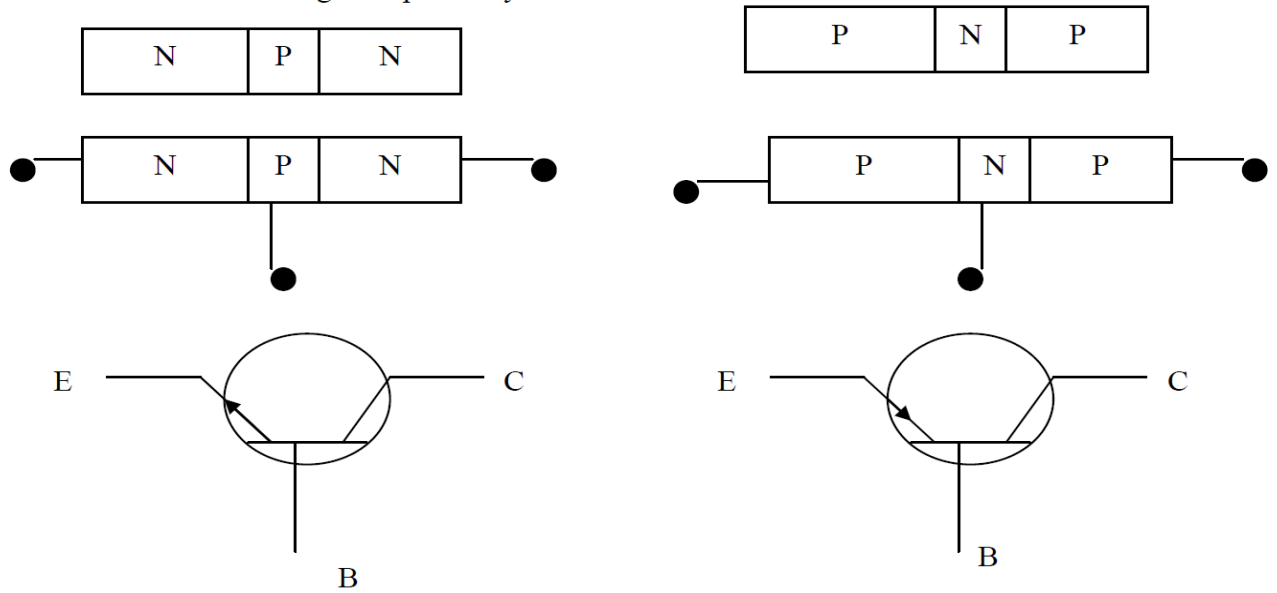


Fig 4.1 Block diagrams and equivalent circuits of (a) NPN and (b) PNP transistors.

A transistor has three terminals, namely emitter (E), base (B) and collector (C). We have two types of transistors, *nnp* and *pnnp*. These are shown in Fig. 2.1.

The emitter is heavily doped and injects a large number of majority carriers into the base. The emitter is always forward biased with respect to the base. In *pnnp* transistors, majority carriers are holes and in *nnp* transistors, majority carriers are electrons.

From Kirchoff's current law,

$$I_E = I_C + I_B \text{ for both PNP and NPN.}$$

2.2.2 *pnp* Transistor

A *pnp* transistor behaves exactly the same way as an *npn* transistor, with the difference that the majority carriers are holes. Here too, the base is lightly doped compared to the emitter and collector. The BE junction is forward biased and CB junction reverse biased as shown in Fig. 2.5.

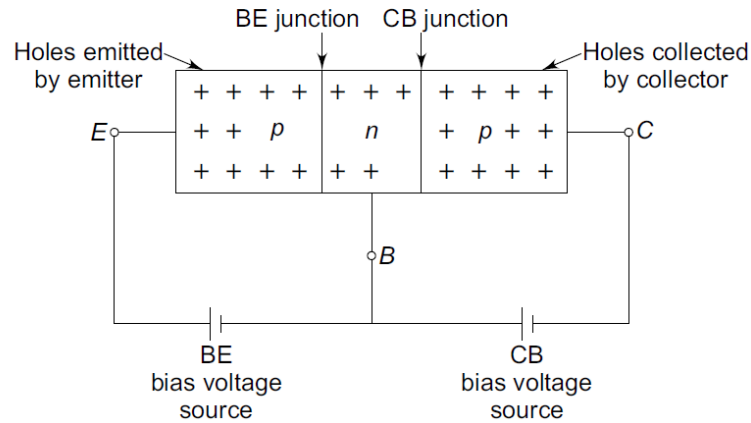


Fig. 2.5 *pnp* transistor

Holes emitted from the *p*-type emitter are injected into the base. Some of the holes flow out through the base. Most of them are collected by the collector. The BE junction bias voltage controls the large emitter and collector current.

Note that the base is biased positive with respect to emitter. In a *pnp* transistor the base is biased negative with respect to emitter. The voltages and source connections are shown in Fig. 2.7.

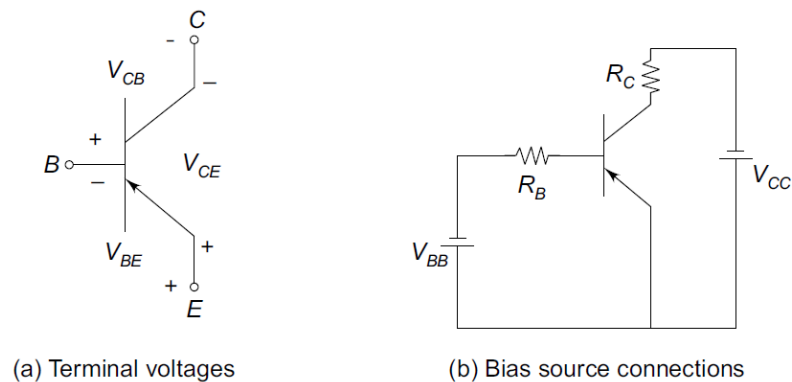


Fig. 2.7 Transistor voltages for *pnp* transistor

BJT Transistors

$$i_E = i_C + i_B \quad i_B = \left(\frac{I_S}{\beta}\right)e^{v_{BE}/V_T} \quad i_E = \frac{\beta+1}{\beta}I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} \quad i_E = \frac{\beta+1}{\beta}i_C$$

the constant β is called the common-emitter **current gain**

$$i_C = \alpha i_E \quad \text{where the constant } \alpha \text{ is related to } \beta \text{ by } \alpha = \frac{\beta}{\beta+1}$$

α is called the **common-base current gain**.

$$i_E = (I_S/\alpha)e^{v_{BE}/V_T} \quad \beta = \frac{\alpha}{1-\alpha}$$

4.2 Common Base (CB) Connections

In common base connection, the base is the ground, i.e all voltages (C-B and E-B) are referenced to the base. The significance of having a common point in a transistor circuit is that it gives a reference for both input voltage to the transistor and the output voltage. In the CB configuration, the E-B voltage is regarded as the 'input voltage' and the C-B as 'output voltage'. In this set-up, E-B voltage is same as emitter-base bias voltage V_{EE} and output voltage (C-B) same as C-B bias voltage (V_{CC}).

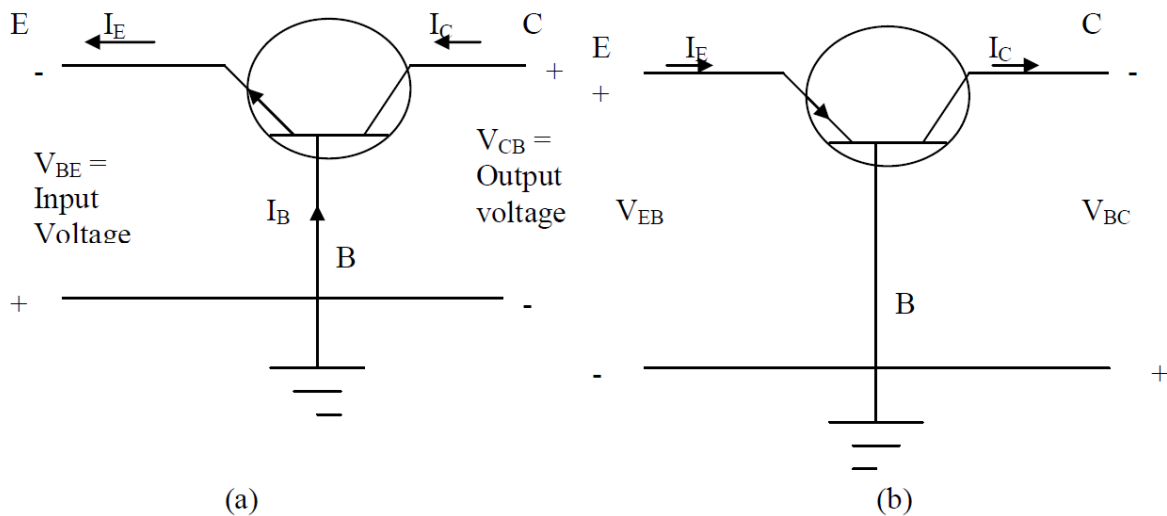


Fig 4.4: Common base connections of (a): NPN and (b) PNP transistors

2.4 TRANSISTOR CURRENTS

Consider the *pnp* transistor with currents as shown in Fig. 2.8.

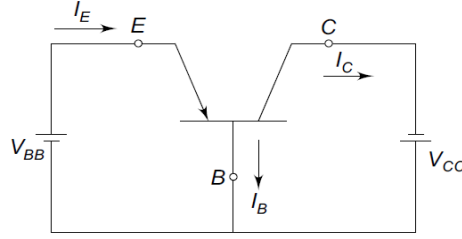


Fig. 2.8 Currents in a *pnp* transistor

The current flowing into the emitter terminal is I_E ; the current flowing out of the base is I_B and out of the collector is I_C . The currents shown are the conventional currents. Here I_B and I_C flow out of the transistor and I_E flows into the transistor. Hence,

$$I_E = I_C + I_B \quad (2.1)$$

Most of the emitter current reaches the collector. Generally, the collector current is around 96% – 99.5% of emitter current. We can write,

$$I_C = \alpha_{dc} I_E \quad (2.2)$$

α_{dc} is the emitter-to-collector current gain and is given by

$$\alpha_{dc} = \frac{I_C}{I_E} \quad (2.3)$$

The value of α lies between 0.96 to 0.995. In circuit analysis often I_C is assumed to be equal to I_E . Due to the collector-base reverse bias, a small reverse saturation current (I_{CBO}) flows across the junction. I_{CBO} is called the collector-to-base leakage current. Substituting (2.1) in (2.2) we get

$$I_C = \alpha_{dc} (I_C + I_B) \quad (2.4)$$

$$I_C(1 - \alpha_{dc}) = \alpha_{dc} I_B$$

$$I_C = \frac{\alpha_{dc}}{1 - \alpha_{dc}} I_B \quad (2.5)$$

$$I_C = \beta_{dc} I_B \quad (2.6)$$

where

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{I_C}{I_B} \quad (2.7)$$

β_{dc} is the base-to-collector current gain. It ranges typically from 25 to 300. From eq. (2.7) we can also obtain

$$\alpha_{dc} = \frac{\beta_{dc}}{1 + \beta_{dc}} \quad (2.8)$$

The currents in an *npn* transistor are shown in Fig. 2.7.

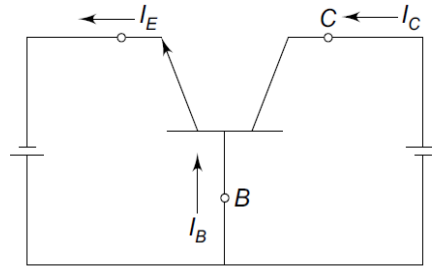


Fig. 2.9 *Currents in an npn transistor*

Here, the base and collector currents enter the transistor and the emitter current leaves the transistor. These are the directions of the conventional currents. All equations from (2.1) to (2.8) are the same.

Example 2.1 Calculate I_C , I_E and I_B for a transistor whose $\alpha_{dc} = 0.9$ and $I_B = 50 \mu\text{A}$. What is β_{dc} ?

Solution

$$\begin{aligned} I_C &= \frac{\alpha_{dc}}{1 - \alpha_{dc}} I_B \\ &= \frac{0.9 \times 50 \times 10^{-6}}{1 - 0.9} = 0.45 \text{ mA} \\ I_E &= \frac{I_C}{\alpha_{dc}} = \frac{0.45 \times 10^{-3}}{0.9} = 0.5 \text{ mA} \\ \beta_{dc} &= \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.9}{1 - 0.9} = 9 \end{aligned}$$

Example 2.2 Calculate I_B , α_{dc} and β_{dc} for transistor that has $I_C = 3 \text{ mA}$ and $I_E = 3.5 \text{ mA}$.

Solution

$$\begin{aligned} \alpha_{dc} &= \frac{I_C}{I_E} = \frac{3 \text{ mA}}{3.5 \text{ mA}} = 0.857 \\ \beta_{dc} &= \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.857}{1 - 0.857} = 5.999 \\ I_B &= \frac{I_C}{\beta_{dc}} = \frac{3 \text{ mA}}{5.999} = 0.5 \text{ mA} \end{aligned}$$

Example 2.3 For a transistor with $\alpha_{dc} = 0.99$, $I_B = 100 \mu\text{A}$, determine I_C

Solution

$$\begin{aligned} \beta_{dc} &= \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.99}{1 - 0.99} = 99 \\ I_C &= \beta_{dc} I_B = 99 \times 100 \times 10^{-6} = 9.9 \text{ mA}. \end{aligned}$$

Example 2.4 Consider a transistor that has a collector current of 3 mA and emitter current of 3.03 mA. Calculate the new currents if the transistor is replaced by a new device that has $\beta = 75$, if the base current is unchanged.

Solution

$$\begin{aligned} \alpha_{dc} &= \frac{I_C}{I_E} = \frac{3 \times 10^{-3}}{3.03 \times 10^{-3}} = 0.99 \\ \beta_{dc} &= \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.99}{1 - 0.99} = 99 \end{aligned}$$

$$I_B = \frac{I_C}{\beta_{\text{dc}}} = \frac{3 \times 10^{-3}}{99} = 30 \text{ } \mu\text{A}$$

If $\beta_{\text{dc}} = 75$,

$$I_C = \beta_{\text{dc}} I_B = 75 \times 30 \text{ } \mu\text{A} = 2.25 \text{ mA}$$

$$\alpha_{\text{dc}} = \frac{\beta_{\text{dc}}}{1 + \beta_{\text{dc}}} = \frac{75}{1 + 75} = 0.98$$

$$I_E = \frac{I_C}{\alpha_{\text{dc}}} = \frac{2.25 \times 10^{-3}}{0.98} = 2.29 \text{ mA}$$